Syllabus for Math 243: Calculus III (Section 001)

Fall 2026 (Aug 24 - Dec 14)

Instructor: Max Hill
Office Hours: 2:00-3:00pm (MW) at Physical Sciences Building 304 (other times available by appointment)
Lectures: 9:30-10:20am (MWF) at WEB 103

Course prerequistes: a grade of C or better in Math 242 or Math 252A. If you don't meet the prerequisites, you'll need to get approval from the math department office.

Course websites:

- Github course page: `https://max-hill.github.io/math-243-F26/`
  - lecture notes will be posted here
  - homework assignments will be posted here
  - exam information
  - syllabus
- Lamaku course page: `https://lamaku.hawaii.edu/d2l/home/170163`
  - course announcements will be posted here
  - you will submit your homework here
  - grades will be posted here
  - (optional) access to the textbook through IDAP

Textbook: *Thomas' Calculus* 15th editiom.
One way to get an electronic version of the textbook is through the *UH Manoa Bookstore's Interactive Digital Access Program* (IDAP), which is going to auto-bill you $35 unless you opt out by September 16. To view the textbook (or opt out), go to the Lamaku course page, click the "Content" tab, then click "VitalSource for UH Manoa". Feel free top opt out of the IDAP if you have other access to the textbook.

Grading (tentative): Homework (10%), Quizzes (20%), Midterms (35%), Final (35%). The following grade cutoffs will be used at the end of the semester to determine final grades:

D	D+	C-	C	C+	B-	B	B+	A-	A	A+
60%	67%	70%	73%	77%	80%	83%	87%	90%	93%	97%

Homework and Quizzes: Both homeworks and quizzes will be approximately weekly.

- Homework is to be turned in on the Lamaku course page. It will be graded on completion. You can use any resources you want, but need to write the homework up yourself to get credit. Late homework will not be accepted.
- In-class quizzes will be approximately weekly and will be based on the homeworks.
- You get one free "no questions asked" make-up quiz (in case you miss a quiz or did poorly on it). This can be done in office hours and doesn't need to be scheduled in advance.

Exams: There will be a midterm exam and a final exam. The final will be cumulative, but will emphasize material after the midterm. The exam dates are as follows:

Midterm exam .... Friday, October 16 (in class)
Final exam .... Monday, December 14, 9:45-11:45am

Make-up policy: Aside from the one "no questions asked" make-up quiz, make-up exams and quizzes are allowed only in three types of circumstances: (1) in accordance with university policies, such as conflict with a religious observation, (2) conflicts with another university-related event, or (3) exceptional circumstances, such as a last-minute medical or family emergency with verification. In the first two cases, notice must be given to the instructor two weeks in advance.

Accommodations: Any student who feels s/he may need an accommodation based on the impact of a disability is invited to contact me privately. I would be happy to work with you, and the KOKUA Program (Office for Students with Disabilities) to ensure reasonable accommodations in my course. KOKUA can be reached at (808) 956-7511 or (808) 956-7612 (voice/text) in Room 013 of the Queen Liliʻuokalani Center for Student Services.

Tentative course outline:

- **Weeks 1-3:** Parametric equations and Polar coordinates
    1. Arc length
    2. Curves defined by parametric equations.
    3. Calculus with plane curves.
    4. Polar coordinates
    5. Areas and lengths in polar coordinates
    6. Conic sections.
- **Weeks 4-6:** Vectors and the geometry of space.
    1. Three-dimensional coordinate systems.
    2. Vectors.
    3. The dot product.
    4. The cross product.
    5. Equations of lines and planes.
    6. Cylinders and quadric surfaces.
- **Friday, October 16 (week 8):** Midterm exam
- **Weeks 7-9:** Vector functions.
    1. Vector functions and space curves
    2. Derivatives and integrals of vector functions.
    3. Arc length and curvature.
    4. Motion in space: velocity and acceleration.
- **Weeks 10-16:** Partial derivatives.
    1. Functions of several variables.
    2. Limits and continuity
    3. Partial derivatives.
    4. Tangent planes and linear approximations
    5. The chain rule.
    6. Directional derivatives and the gradient vector.
    7. Maximum and minimum values.
    8. Discovery project (p.1010 Stewart): Quadratic approximation and critical points (Taylor's formula for two variables).
    9. Lagrange multipliers.
- **Monday, December 14:** Final exam.